

The recovery methods workshop

Teacher guide and worked answers

Choose and justify filtration, evaporation or evaporation with condensation for a stated recovery target.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Summer | C34

Investigate recovering substances from reversible changes.

Intentional improved revisit: historical GROW Solubility_And_Recovery/Getting_The_Solid_Back lessons cover related content. Original secondary-age exemplar at an upper-KS2 conceptual level; no qualification approval or completion claim.

Prior learning

Distinguish dissolving from melting.

Know that a solution contains a dissolved substance and a solvent.

Success evidence

Identify the mixture and desired recovered substance.

Explain what passes through a filter and what remains.

Sequence a recovery route and account for each component.

Equipment

Shared mixture/target cards, selected route, pencils; optional sequencing tokens.

Preparation

Default lesson is a process simulation and paper investigation. No heated separation apparatus is used.

Name the target before choosing a method. The model follows sand, salt and water; it does not certify drinking-water purity.

Practical care

Do not heat, boil, distil or taste any classroom mixture as part of this pack.

The evaporation/condensation diagram is a schematic of processes, not an apparatus construction guide.

Optional physical filtration would require a separately prepared school method; all necessary evidence is already supplied.

Ready to teach alternative

Use the four recovery briefs and process cards. Screen process choices and paper sequencing both provide the complete investigation.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Same mixture, different mission
2	Retrieval	2-5	Recover the mixture ideas
3	I do	5-13	I decide what the filter can retain
4	I do	Within stage	I target the dissolved solid
5	I do	Within stage	I change the plan when the target changes
6	We do	13-21	Recovery workshop: choose, run, explain
7	We do	Within stage	Two targets need a sequence
8	We do	Within stage	Repair the purity claim
9	Hinge	21-24	A team wants water from saltwater
10	You do	24-34	Your independent investigation
11	You do	Within stage	Use the evidence
12	You do	Within stage	Check before sharing
13	Review	34-38	Compare and improve
14	Review	Within stage	A question worth investigating
15	Exit	38-40	Show the science you can explain

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Same mixture, different mission

Start with target choice. The learner controls a process model, not real heating equipment.

2 Retrieval Recover the mixture ideas

Yes. Evaporation changes liquid to gas at a surface; condensation changes gas to liquid. Evaporation can occur below the boiling temperature.

3 I do I decide what the filter can retain

Think aloud: I look at solubility, not just appearance. The ordinary filter catches sand grains but does not remove dissolved salt. The diagram is schematic; blue liquid below still contains salt.

4 I do I target the dissolved solid

Shared I do time. Model the target statement first: I need salt. Then trace the water leaving and salt remaining. No heating is requested; evaporation can occur over time at room temperature.

5 I do I change the plan when the target changes

Shared I do time. The combined processes are the basis of distillation; naming distillation is optional. This is not a build diagram and does not certify drinking-water safety.

6 We do Recovery workshop: choose, run, explain

A: filter to collect sand. B: evaporate water to leave salt. C: capture water vapour and condense it to collect water. Ask pupils to account for both components.

7 We do Two targets need a sequence

Filter first: sand remains as residue and saltwater passes through as filtrate. Evaporate water from the filtrate to collect salt. Sand may need washing and drying for cleaner recovery; added wash water must be included when accounting for the mixture.

8 We do Repair the purity claim

Ordinary filtration does not reliably remove dissolved substances or microbes. Do not drink the mixtures. A clear liquid is not proven pure or safe.

Reveal: Ordinary filtration can leave dissolved substances in the liquid. Appearance does not prove purity or safety.

9 Hinge A team wants water from saltwater

A chooses the wrong target. B correctly captures and condenses water vapour. C fails because ordinary filtration does not remove dissolved salt.

A. Let water evaporate into the room and keep the salt. - That recovers salt while losing the target water.

B. Capture water vapour and condense it into a collecting container. - The target water is collected again as liquid.

C. Filter once and assume all dissolved salt is trapped. - Dissolved salt passes through ordinary filter paper.

10 You do Your independent investigation

10 minutes across the three You do slides. Supply shared page 1 and one selected route page. All routes target the stated science objective; a labelled drawing or independent dictated explanation is valid.

11 You do Use the evidence

Shared You do time. Ask pupils which evidence supports their explanation. Read prompts or scribe without supplying the scientific conclusion.

12 You do Check before sharing

Shared You do time. Use the route-specific worked answers to identify one precise next step.

13 Review Compare and improve

4 minutes across Review. Offer partner or private teacher feedback. Ask for one specific revision linked to evidence; no public ranking.

14 Review A question worth investigating

Lundy cycle: invite written, spoken or pointed suggestions. Select one feasible question and explain how the pupil contribution changes the next example or investigation.

15 Exit Show the science you can explain

Insoluble sand is caught by the filter. Removing water by evaporation leaves salt. Capture vapour and condense it to collect water. E4 asks why clear filtered liquid is not necessarily pure or safe.

Answers and assessment

R1-R3

Dissolved salt remains present. Evaporation changes liquid to gas at its surface; it can occur below boiling. Condensation changes gas to liquid.

W1-W3

A filtration gives sand residue; B evaporation leaves salt; C capture vapour and condense to collect water. D filter first, then evaporate saltwater filtrate; sand stays as residue. W3 dissolved substances may pass through and other contaminants may remain; clarity is not purity or drinking-water safety.

Hinge A-C

A loses the water target. B correct. C ordinary filtration does not trap dissolved salt.

S1-S6

Filter; evaporate water; capture and condense; sand residue remains while saltwater passes; filter then evaporate filtrate; no, clarity alone is insufficient.

N1-N4

A filter/sand; B evaporate/salt; C evaporate with capture and condensation/water; D filter then evaporate filtrate/sand and salt separately. N2 dissolved salt passes with the water through ordinary filter paper. N3 vapour escapes unless captured. N4 sand residue and saltwater filtrate; remove water from filtrate to leave salt.

T1-T4

T1 salt target: water leaves and salt remains; water target: capture and condense water vapour while salt remains at source. T2 filter sand before removing water from dissolved salt; evaporating first leaves mixed sand and salt rather than separate products. T3 sample may remain dissolved in unrecovered liquid, stick to apparatus/filter, spill, or be affected by measurement uncertainty. Missing measured mass is not proof of destruction. T4 filtration establishes separation of suitable insoluble particles, not all dissolved substances or microbes; no safety claim is justified.

E1-E4 / assessment

Insoluble sand; remove water by evaporation; capture and condense vapour; dissolved contaminants can remain and appearance is insufficient. Secure: target-dependent route and component accounting. Re-teach method matching by asking where the desired component finishes. Reflection accepts a reasoned target comparison.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

This is an improved application of earlier solubility and recovery work. The new focus is target-dependent selection, component accounting and a multistep mixture.

The interactive separation model uses exactly briefs A-D on paper. It animates a process sequence, not a complete apparatus or practical safety procedure.

Evaporation is not synonymous with boiling. Recovering water uses captured vapour and condensation; no distillation apparatus is built in this lesson.

Access and responsive teaching

Supported

Match mixture/target pairs to process cards.

Standard

Explain four routes and account for components.

Stretch

Evaluate multistep recovery, loss and limits of purity claims.

Regulation

Offer private writing, pointing, drawing or adult scribing. Allow pass-and-return for public questions and desk-based participation. Choose support from current learning evidence, not fixed ability labels.

Misconceptions to check

A filter removes dissolved salt from water.

Dissolved salt passes with water through ordinary filter paper; sand can be retained.

Check: Ask where the salt appears after filtering.

Evaporation automatically gives collected liquid water.

Open evaporation lets water vapour escape; collecting water requires capture and condensation.

Check: Ask where the target water ends up.

One separation method works for every mixture and target.

The method depends on component properties and what you want to collect.

Check: Compare saltwater target salt with saltwater target water.

Word	Meaning
filtration	Separating an insoluble solid from a fluid using a filter.
residue	Material left on the filter.
filtrate	Fluid that passes through a filter.
evaporation	Liquid changing to gas at its surface.
condensation	Gas changing to liquid.
recovery target	The substance or substances that a method is intended to collect.

Scientific references

[Royal Society of Chemistry: Separation techniques](#)

Checked September 2026. Filtration, evaporation and separation misconceptions. Classroom activities are original and include a complete paper route.

[Royal Society of Chemistry: Separating sand and salt](#)

Checked September 2026 through primary search extract. Solubility, filtration and evaporation; this pack does not reproduce the heated practical.

Royal Society of Chemistry: Evaporation, filtration and crystallisation

Checked September 2026. Separation teaching and target-dependent methods. This resource supplies a non-heated paper alternative.